FINITE-DIMENSIONAL SPACES WHOSE DUAL BALL HAS COUNTABLY MANY EXTREME POINTS HAVE LINDENSTRAUSS PROPERTY B

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Abstract. We prove that every finite-dimensional real Banach space Y whose dual unit ball has only countably many extreme points has Lindenstrauss property B: for every Banach space X, the norm attaining operators are dense in $\mathcal{L}(X, Y)$. This strictly extends the polyhedral case (Lindenstrauss) and the case of property quasi-β (Acosta, Aguirre and Payá); in dimension two it is, as far as we know, the first extension beyond polyhedral spaces. As a consequence, every finite-dimensional subspace of $C(K)$, for a countable compact Hausdorff space K, has property B; in particular every finite-dimensional subspace of c has property B, which answers a question of Jung, Martín and Rueda Zoca. We introduce property scattered quasi-β, which allows the uniformity in property quasi-β to fail on a scattered set of extreme points, and show that for finite-dimensional spaces it is equivalent to the dual ball having countably many extreme points. The proof combines the Bourgain–Stegall theorem, a dichotomy expressed in terms of Birkhoff–James orthogonality at the peak of the adjoint, and a descent along the Cantor–Bendixson levels of the set of extreme points.

Status of this manuscript. This manuscript was written by an AI model (CLAUDE) under the guidance of Debmalya Sain. It is not meant for publication. At the time of writing, its results have *not yet been verified by a human*. Every proof has been checked by the author, but the reader should treat the results, and in particular the answer to the question of Jung, Martín and Rueda Zoca, as unrefereed until a human verification is recorded here. The contributions of the witness and steerer are described in detail in Section 6.

1. Introduction

Throughout, all Banach spaces are real. For Banach spaces X, Y, an operator $T \in \mathcal{L}(X, Y)$ *attains its norm* if $\|Tx_0\| = \|T\|$ for some x_0 in the unit ball B_X. The set of norm attaining operators is denoted by $\mathrm{NA}(X, Y)$. Following Lindenstrauss [9], Y has *property B* if $\mathrm{NA}(X, Y)$ is dense in $\mathcal{L}(X, Y)$ for every Banach space X. Bishop and Phelps [3] proved that $\mathbb{R}$ has property B.

Lindenstrauss introduced property β and showed that it implies property B; moreover, finite-dimensional polyhedral spaces have property β [9]. Acosta, Aguirre and Payá [1] introduced the weaker property quasi-β, which also implies property B. For a finite-dimensional space Y, property quasi-β holds exactly when $\operatorname{ext} B_{Y^*}$ is discrete; this was observed in [1, p. 414] (see [6, Remark 3.19]), and we include a short proof in Corollary 5.7. A discrete subset of the compact metric space B_{Y^*} is countable, so every finite-dimensional space with property quasi-β satisfies the hypothesis of Theorem 1.1 below. In dimension two, $\operatorname{ext} B_{Y^*}$ is closed, hence compact, so it is discrete only when it is finite; thus for two-dimensional spaces, property β, property quasi-β and polyhedrality coincide (see [6, Remark 3.19] and Remark 5.12).

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Whether every finite-dimensional space has property B is a well-known open problem [7, 10]. Jung, Martín and Rueda Zoca observed that c contains a two-dimensional subspace whose dual ball has infinitely many extreme points, so that this subspace fails property quasi-β, and remarked that it is not known whether such a subspace has property B [6, Remark 3.24]. They also obtained results for finite-dimensional spaces Y such that $\operatorname{ext} B_{Y^*}$ is countable, under additional assumptions on the domain X [6, Example 3.18].

Our main result removes every assumption on X.

Theorem 1.1. *Let Y be a finite-dimensional real Banach space such that $\operatorname{ext} B_{Y^*}$ is countable. Then Y has property B.*

The subspace of c mentioned above shows that Theorem 1.1 strictly extends the case of property quasi-β, already in dimension two. More generally:

Corollary 1.2. *Let K be a countable compact Hausdorff space. Then every finite-dimensional subspace of $C(K)$ has property B. In particular, every finite-dimensional subspace of c has property B.*

Proof. Let $Y \subseteq C(K)$ be finite-dimensional, and for $k \in K$ let $\delta_k \in C(K)^*$ be the evaluation at k. Since $\|f\| = \sup_{k\in K} |\delta_k(f)|$ for $f \in C(K)$, the bipolar theorem shows that $B_{C(K)^*}$ is the weak*-closed absolutely convex hull of $\{\delta_k : k \in K\}$. Let $\varrho\colon C(K)^* \to Y^*$ be the restriction map. By the Hahn–Banach theorem, $\varrho(B_{C(K)^*}) = B_{Y^*}$. Since Y is finite-dimensional, ϱ is continuous from the weak* topology of $C(K)^*$ to the norm topology of Y^*. Therefore B_{Y^*} is contained in the closure of the absolutely convex hull of

$$D = \{\delta_k|_Y : k \in K\},$$

and hence B_{Y^*} is the closed convex hull of $D \cup -D$. For every $y \in Y$ the function $k \mapsto \delta_k|_Y(y) = y(k)$ is continuous on K; since Y is finite-dimensional, it follows that $k \mapsto \delta_k|_Y$ is continuous from K into Y^* (it suffices to check this on a basis of Y). So $D \cup -D$ is compact, and it is countable because K is. By Milman's converse to the Krein–Milman theorem [11], $\operatorname{ext} B_{Y^*} \subseteq D \cup -D$, which is countable. Now apply Theorem 1.1. Finally, c is isometric to $C(\omega + 1)$, where $\omega + 1 = \mathbb{N} \cup \{\infty\}$ is the one-point compactification of $\mathbb{N}$, via $(\xi_n) \mapsto (n \mapsto \xi_n,\ \infty \mapsto \lim_n \xi_n)$. □

The proof of Theorem 1.1 works entirely with the adjoint $S^*\colon Y^* \to X^*$. After a small perturbation, S^* attains its norm on B_{Y^*} only at a pair $\pm p$ of extreme points (Bourgain–Stegall). The decisive question is whether p is Birkhoff–James orthogonal to a direction from which other extreme points approach it, with the orthogonality witnessed at a landing point of S (Definition 3.2 and Remark 3.3). If it is not (*steep* case), a rank-one nudge using the Bishop–Phelps theorem produces a norm attaining operator. If it is (*flat* case), the orthogonality is preserved by S^* (Lemma 2.5), and a small tilt raises a nearby extreme point above p. The new peak has strictly smaller Cantor–Bendixson level, so the process stops after finitely many steps.

In Section 5 we place Theorem 1.1 in the framework of property quasi-β. We introduce *property scattered quasi-β*, which keeps the conditions of Acosta, Aguirre and Payá but allows the uniformity near extreme points to fail on a scattered set of *bad* extreme points. For finite-dimensional spaces it is equivalent to the dual ball having countably many extreme points (Proposition 5.6), so Theorem 1.1 can be read as: *every finite-dimensional space with property scattered quasi-β has property B* (Corollary 5.8).

Section 6 describes the contributions of the witness and steerer, and Section 7 describes how this manuscript was produced and what remains to be verified.

2. Tools

2.1. **Notation.** Throughout Sections 2–4, X is an arbitrary real Banach space and Y is a finite-dimensional real Banach space. For a subset F of a normed space, $\operatorname{dist}(\cdot, F)$ denotes the distance to F. For $x^* \in X^*$ and $y \in Y$, $x^* \otimes y$ is the operator $x \mapsto x^*(x)y$. We write

$$K_Y = B_{Y^*}, \qquad E = \operatorname{ext} K_Y.$$

Since Y is finite-dimensional we identify $Y^{**} = Y$; in particular the weak* and norm topologies of Y^* coincide, K_Y is compact, and $K_Y = \operatorname{conv} E$ by Minkowski's theorem. Elements of K_Y are denoted by h; elements of E near a peak p are denoted by u; w denotes points of the landing set W_S (Section 2.3); τ denotes approach directions (Section 3); λ denotes Cantor–Bendixson levels. Given x, v in a normed space, $x \perp_B v$ means that $\|x + \xi v\| \geq \|x\|$ for every $\xi \in \mathbb{R}$ (Birkhoff–James orthogonality [2, 5]).

2.2. **Unique peaks.** An operator $R \in \mathcal{L}(Z, V)$ *absolutely strongly exposes* B_Z if there is $\hat{b} \in B_Z$ such that every sequence $(b_n) \subseteq B_Z$ with $\|Rb_n\| \to \|R\|$ has a subsequence converging to $\hat{b}$ or to $-\hat{b}$.

Proposition 2.1 (Bourgain–Stegall; see [6, Proposition 1.1(a)])**.** *Let Z have the Radon–Nikodým property and let V be a Banach space. For every $R \in \mathcal{L}(Z, V)$ and $\varepsilon > 0$ there are $z^* \in S_{Z^*}$, $v \in S_V$ and $0 < \theta < \varepsilon$ such that $R + \theta\, z^* \otimes v$ absolutely strongly exposes B_Z.*

This is due to Bourgain [4], and the rank-one form follows from Stegall's variational principle [13].

Definition 2.2. Let $S \in \mathcal{L}(X, Y)$, $S \neq 0$, and $p \in K_Y$. We say that S has *peak* $\pm p$ if $\|S^*p\| = \|S\|$ and, for every $\delta > 0$, there is $\gamma(\delta) > 0$ such that

$$\|S^*h\| \leq \|S\| - \gamma(\delta) \qquad \text{whenever } h \in K_Y,\ \operatorname{dist}(h, \{p, -p\}) \geq \delta. \tag{1}$$

If S has peak $\pm p$, then $\pm p$ are the only points of K_Y at which S^* attains its norm, and $p \in E$. Indeed, if $p = \frac{1}{2}(h_1 + h_2)$ with $h_1, h_2 \in K_Y$, then $\|S\| = \|S^*p\| \leq \frac{1}{2}(\|S^*h_1\| + \|S^*h_2\|) \leq \|S\|$, so S^* attains its norm at h_1 and at h_2. Hence $h_1, h_2 \in \{p, -p\}$, and since $p \neq 0$ (because $\|S^*p\| = \|S\| > 0$), this forces $h_1 = h_2 = p$. Moreover $\|p\| = 1$, since $\|S\| = \|S^*p\| \leq \|S\|\|p\|$. Clearly, S has peak $\pm p$ if and only if cS has peak $\pm p$, for every $c > 0$.

Corollary 2.3. *Let $T \in \mathcal{L}(X, Y)$, $T \neq 0$, and $0 < \varepsilon < \|T\|$. There is $S \in \mathcal{L}(X, Y)$ with $\|S - T\| < \varepsilon$ which has a peak.*

Proof. Apply Proposition 2.1 with $Z = Y^*$ (finite-dimensional, hence with the Radon–Nikodým property), $V = X^*$ and $R = T^*$. Here $z^* = y_0 \in S_Y$, and $T^* + \theta\, y_0 \otimes v$ is the adjoint of $S = T + \theta\, v \otimes y_0$, so $\|S - T\| = \theta < \varepsilon$; in particular $S \neq 0$. Let $p = \hat{b}$ be the point given by absolute strong exposition. A norming sequence for S^* in the compact set K_Y has a subsequence converging to p or to $-p$, so $\|S^*p\| = \|S\|$. If (1) failed for some $\delta > 0$, there would be $h_n \in K_Y$ with $\operatorname{dist}(h_n, \{p, -p\}) \geq \delta$ and $\|S^*h_n\| \to \|S\|$; a subsequence would converge to p or to $-p$, which is absurd. □

2.3. **The floor.** For $S \in \mathcal{L}(X, Y)$ with $\|S\| = 1$ and peak $\pm p$, the *landing set* is

$$W_S = \{w \in \operatorname{cl} S(B_X) : p(w) = 1\}.$$

It is compact, since $\operatorname{cl} S(B_X) \subseteq B_Y$ is compact, and it is nonempty: if $x_n \in B_X$ and $p(Sx_n) \to 1$ (such x_n exist because $\sup_{x \in B_X} p(Sx) = \|S^*p\| = 1$), every limit point of (Sx_n) lies in W_S.

Lemma 2.4 (Floor)**.** *If $\|S\| = 1$, then $\|S^*h\| \geq h(z)$ for all $h \in Y^*$ and all $z \in \operatorname{cl} S(B_X)$. In particular, this holds for $z \in W_S$.*

Proof. $\|S^*h\| \geq (S^*h)(x) = h(Sx)$ for $x \in B_X$, and the right-hand side is continuous in Sx. □

The floor is elementary. Its role in this article is to transport Birkhoff–James orthogonality at the peak, as follows.

Lemma 2.5 (Preservation of orthogonality at the peak)**.** *Let $\|S\| = 1$ with peak $\pm p$, let $w \in W_S$, and let $\tau \in Y^*$ satisfy $\tau(w) = 0$. Then $p \perp_B \tau$ in Y^*, and $S^*p \perp_B S^*\tau$ in X^*.*

Proof. For every $\xi \in \mathbb{R}$, since $\|w\| \leq 1$ and $p(w) = 1$,

$$\|p + \xi\tau\| \geq (p + \xi\tau)(w) = 1 = \|p\|,$$

and, by Lemma 2.4,

$$\|S^*p + \xi S^*\tau\| = \|S^*(p + \xi\tau)\| \geq (p + \xi\tau)(w) = 1 = \|S^*p\|. \qquad \square$$

Remark 2.6. Lemma 2.5 says that the operator S^*, which attains its norm at p, preserves those Birkhoff–James orthogonalities at p which are witnessed at a landing point. This is the phenomenon studied in [12, Theorem 2.3], where it is shown that an operator on a smooth Banach space preserves Birkhoff–James orthogonality at every point at which it attains its norm. That theorem was the conceptual source of Lemmas 2.4 and 2.5. It is not used as a logical step in this article: our setting is not covered by it, since S^* acts between two different spaces and Y^* need not be smooth, and the elementary Lemma 2.4 suffices.

2.4. **Levels.**

Lemma 2.7. *If E is countable, then E is scattered. Consequently every $q \in E$ has an ordinal level $\lambda(q)$ such that q has a neighbourhood in K_Y containing no point $q' \in E \setminus \{q\}$ with $\lambda(q') \geq \lambda(q)$, and $\lambda(-q) = \lambda(q)$.*

Proof. The set of non-extreme points of K_Y is $\bigcup_n F_n$, where $F_n = \{\frac{1}{2}(h_1 + h_2) : h_1, h_2 \in K_Y,\ \|h_1 - h_2\| \geq 1/n\}$ is compact. So E is a G_δ in the compact metric space K_Y, hence a Polish space. Every nonempty closed subset of E is a countable Polish space, so by the Baire category theorem it has an isolated point. Thus E is scattered. Let $E^{(0)} = E$, let $E^{(\alpha+1)}$ be the set of non-isolated points of $E^{(\alpha)}$, and take intersections at limit ordinals. These Cantor–Bendixson derivatives vanish at some countable ordinal [8]. Define $\lambda(q)$ as the unique α with $q \in E^{(\alpha)} \setminus E^{(\alpha+1)}$. Then q is isolated in $E^{(\lambda(q))} = \{q' \in E : \lambda(q') \geq \lambda(q)\}$, which is the claim. Symmetry follows since $h \mapsto -h$ is a homeomorphism of E onto itself. □

Remark 2.8. Conversely, a scattered subset of the separable metric space K_Y is countable (see the proof of Proposition 5.6). So for finite-dimensional Y, the set E is countable if and only if it is scattered, and Theorem 1.1 can equally be stated for spaces whose dual ball has a scattered set of extreme points.

3. Flat and steep peaks

Let $\|S\| = 1$ with peak $\pm p$, and assume p is not isolated in E. Let $\mathcal{T} \subseteq S_{Y^*}$ be the set of limits of $(u_k - p)/\|u_k - p\|$ along sequences $u_k \in E \setminus \{p\}$ with $u_k \to p$. Since p is not isolated and S_{Y^*} is compact, $\mathcal{T}$ is nonempty; a diagonal argument shows that $\mathcal{T}$ is closed, hence compact. We call the elements of $\mathcal{T}$ the *approach directions* at p.

Lemma 3.1. *$\tau(w) \leq 0$ for all $\tau \in \mathcal{T}$ and $w \in W_S$.*

Proof. For $u \in E$ and $w \in W_S$, Lemma 2.4 gives $u(w) \leq \|S^*u\| \leq 1 = p(w)$. So $(u - p)(w) \leq 0$. Divide by $\|u - p\|$ and pass to the limit. □

Definition 3.2. The peak p is *flat* if $\tau(w_0) = 0$ for some $\tau \in \mathcal{T}$ and some $w_0 \in W_S$. Otherwise, by compactness of $\mathcal{T}$ and W_S and Lemma 3.1, there is $\nu > 0$ with $\tau(w) \le -2\nu$ for all $\tau \in \mathcal{T}$ and $w \in W_S$, and p is *steep* with constant ν.

Remark 3.3. By James' characterisation [5], applied in Y^* with $Y^{**} = Y$, $p \perp_B \tau$ if and only if $\tau(w) = 0$ for some w in the face $M_p = \{w \in B_Y : p(w) = 1\}$. Since $W_S \subseteq M_p$, a flat peak is Birkhoff–James orthogonal to an approach direction, the orthogonality is witnessed at a landing point of S, and by Lemma 2.5 it is preserved by S^*. The converse may fail: a steep peak can still be Birkhoff–James orthogonal to an approach direction, through a point of $M_p \setminus W_S$. This is why flatness is defined relative to W_S.

Lemma 3.4 (Uniform steepness)**.** *Let p be steep with constant ν. There is $\delta_1 > 0$ such that*

$$(u - p)(z) \le -\nu\|u - p\|$$

for every $u \in E$ with $0 < \|u - p\| < \delta_1$ and every $z \in B_Y$ with $\operatorname{dist}(z, W_S) < \nu/2$.

Proof. Write $v = (u - p)/\|u - p\|$. There is $\delta_1 > 0$ such that $\operatorname{dist}(v, \mathcal{T}) < \nu/2$ whenever $u \in E$ and $0 < \|u - p\| < \delta_1$. Otherwise a sequence $u_k \to p$ in $E \setminus \{p\}$ would have directions at distance at least $\nu/2$ from $\mathcal{T}$, and by compactness of S_{Y^*} a limit direction outside $\mathcal{T}$, which is absurd. Choose $\tau \in \mathcal{T}$ with $\|v - \tau\| < \nu/2$ and $w \in W_S$ with $\|z - w\| < \nu/2$. Since $\|z\| \le 1$ and $\|\tau\| = 1$,

$$v(z) = \tau(w) + (v - \tau)(z) + \tau(z - w) < -2\nu + \nu/2 + \nu/2 = -\nu. \qquad \square$$

4. The two steps

Proposition 4.1 (Steep or isolated peaks)**.** *Let $\|S\| = 1$ with peak $\pm p$, where p is isolated in E or steep. Then for every $\varepsilon > 0$ there is $S' \in \mathrm{NA}(X, Y)$ with $\|S' - S\| < \varepsilon$.*

Proof. Constants. These depend only on S and ε, and are fixed before the nudge.

Case 1: p is isolated in E. Choose $0 < \delta_2 < 1$ such that no point of $E \setminus \{p, -p\}$ lies within δ_2 of p or of $-p$. Let $\gamma = \gamma(\delta_2)$ be as in (1), and choose $0 < r < \min\{\varepsilon,\ \gamma/2\}$.

Case 2: p is not isolated and is steep with constant ν. Let δ_1 be as in Lemma 3.4. Let

$$1 - \kappa = \max\{p(z) : z \in \operatorname{cl} S(B_X),\ \operatorname{dist}(z, W_S) \ge \nu/2\}$$

(take $\kappa = 1$ if the set is empty). Then $\kappa > 0$, since $p(z) = 1$ with $z \in \operatorname{cl} S(B_X)$ forces $z \in W_S$. Put $\delta_2 = \min\{\delta_1, \kappa/4, 1/2\}$, let $\gamma = \gamma(\delta_2)$ be as in (1), and choose

$$0 < r < \min\{\varepsilon,\ \gamma/2,\ \nu,\ \kappa/8\}.$$

The nudge. Fix $m \in W_S$; then $\|m\| \le 1 = p(m)$. By the Bishop–Phelps theorem [3] choose $\varphi \in \mathrm{NA}(X, \mathbb{R})$ with $\|\varphi - S^*p\| < r$, and put

$$S' = S + (\varphi - S^*p) \otimes m.$$

Then $\|S' - S\| < r$, $S'^*h = S^*h + h(m)(\varphi - S^*p)$ for $h \in Y^*$, $S'^*p = \varphi$, and $\|\varphi\| \ge 1 - r$.

Claim: $\|S'^*h\| \le \|\varphi\|$ for every $h \in E$. Since $h \mapsto \|S'^*h\|$ is convex and $K_Y = \operatorname{conv} E$, the claim gives $\|S'\| = \|S'^*\| = \|S'^*p\| = \|\varphi\|$. If φ attains its norm at $x_0 \in B_X$, say $\varphi(x_0) = \|\varphi\|$, then $\|S'x_0\| \ge p(S'x_0) = (S'^*p)(x_0) = \varphi(x_0) = \|S'\|$, so $S' \in \mathrm{NA}(X, Y)$.

Since $\|S'^*(-h)\| = \|S'^*h\|$, it suffices to treat $h \in E$ with $\operatorname{dist}(h, \{p, -p\}) \ge \delta_2$, $h = p$, and $h = u \in E$ with $0 < \|u - p\| < \delta_2$.

(i) If $\operatorname{dist}(h, \{\pm p\}) \ge \delta_2$, then $\|S'^*h\| \le \|S^*h\| + r \le 1 - \gamma + r \le 1 - r \le \|\varphi\|$.

(ii) If $h = p$, then $S'^*p = \varphi$.

(iii) Let $u \in E$ with $0 < \|u - p\| < \delta_2$. This case occurs only in Case 2. Since $p(m) = 1$,

$$S'^*u = \varphi + S^*(u - p) + (u - p)(m)\,(\varphi - S^*p).$$

Let $x \in B_X$ and $z = Sx \in B_Y$. If $\operatorname{dist}(z, W_S) < \nu/2$, Lemma 3.4 and $|(u-p)(m)| \le \|u-p\|$ give

$$(S'^* u)(x) \le \|\varphi\| - \nu\|u-p\| + r\|u-p\| \le \|\varphi\|.$$

If $\operatorname{dist}(z, W_S) \ge \nu/2$, then $p(z) \le 1-\kappa$, and

$$(S'^* u)(x) = u(z) + u(m)(\varphi - S^*p)(x) \le p(z) + \|u-p\| + r \le 1 - \kappa + \kappa/4 + \kappa/8 < 1 - r \le \|\varphi\|.$$

Taking the supremum over $x \in B_X$ (which is symmetric) gives $\|S'^* u\| \le \|\varphi\|$. □

Proposition 4.2 (Flat peaks: tilt and descend)**.** *Let E be countable. Let $\|S\| = 1$ with peak $\pm p$, where p is not isolated in E and is flat. Then for every $\varepsilon > 0$ there is S_{peak} with $\|S_{\text{peak}} - S\| < \varepsilon$ which has a peak $\pm p'$ with $\lambda(p') < \lambda(p)$.*

Proof. Let $\tau \in \mathcal{T}$ and $w_0 \in W_S$ with $\tau(w_0) = 0$. Since $p(w_0) = 1$ and $\|\tau\| = 1$, τ is not a multiple of p, so there is $d \in Y$ with $p(d) = 0$ and $\tau(d) = 1$. By Lemma 2.7 choose $0 < \delta < 1$ such that no point $q \in E \setminus \{p, -p\}$ with $\lambda(q) \ge \lambda(p)$ lies within δ of p or of $-p$. Let $\gamma = \gamma(\delta)$ and choose $s > 0$ with $s\|d\| < \min\{\gamma, \varepsilon/2\}$. Define

$$S_{\text{tilt}} x = Sx + s\, p(Sx)\, d, \qquad \text{so that} \qquad S^*_{\text{tilt}} h = S^*\big(h + s\, h(d)\, p\big).$$

Then $\|S_{\text{tilt}} - S\| \le s\|d\|$ and $\|S^*_{\text{tilt}}(\pm p)\| = \|S^* p\| = 1$, because $p(d) = 0$.

The tilt raises a neighbour. Let $u_k \in E \setminus \{p\}$ with $u_k \to p$ and $v_k = (u_k - p)/\|u_k - p\| \to \tau$. By Lemma 2.4 applied to S at $w_0 \in W_S$, and since $p(w_0) = 1$ and $p(d) = 0$,

$$\|S^*_{\text{tilt}} u_k\| \ge \big(u_k + s\, u_k(d)\, p\big)(w_0) = 1 + \|u_k - p\|\big(v_k(w_0) + s\, v_k(d)\big).$$

As $k \to \infty$, $v_k(w_0) + s\, v_k(d) \to \tau(w_0) + s\,\tau(d) = s > 0$. So $\|S_{\text{tilt}}\| = 1 + \mu$ for some $\mu > 0$.

Far points stay low. If $h \in K_Y$ and $\operatorname{dist}(h, \{\pm p\}) \ge \delta$, then $\|S^*_{\text{tilt}} h\| \le \|S^* h\| + s\|d\| \le 1 - \gamma + s\|d\| < 1$.

Re-peaking. By Corollary 2.3 there is S_{peak} with $\|S_{\text{peak}} - S_{\text{tilt}}\| = \theta' < \min\{\mu/2, \varepsilon/2\}$ which has a peak $\pm p'$. Then $\|S_{\text{peak}}\| > 1 + \mu/2$, while $\|S^*_{\text{peak}} h\| < 1 + \mu/2$ both for $h = \pm p$ and for $h \in K_Y$ at distance at least δ from $\pm p$. Hence $p' \notin \{p, -p\}$ and, after replacing p' by $-p'$ if necessary, $\|p' - p\| < \delta$. Since $p' \in E$, the choice of δ gives $\lambda(p') < \lambda(p)$. Finally $\|S_{\text{peak}} - S\| < \varepsilon$. □

Proof of Theorem 1.1. Let $T \in \mathcal{L}(X, Y)$, $T \ne 0$, and $\varepsilon > 0$. It suffices to treat $\varepsilon < \|T\|$. Put $L = \|T\| + \varepsilon$. Peaks, levels, flatness and membership in $\mathrm{NA}(X, Y)$ are unchanged when an operator is multiplied by a positive scalar (flatness and steepness of a peak of $S \ne 0$ are defined through $S/\|S\|$). So Propositions 4.1 and 4.2 apply to any $S \ne 0$ with peak $\pm p$: apply them to $S/\|S\|$ with tolerance η/L and multiply back by $\|S\| \le L$, which gives a perturbation of norm less than η. By Corollary 2.3 choose S_0 with $\|S_0 - T\| < \varepsilon/2$ which has a peak $\pm p_0$. As long as the current peak p_k is non-isolated and flat, apply Proposition 4.2 in this way with $\eta = \varepsilon/2^{k+2}$, obtaining S_{k+1} with peak $\pm p_{k+1}$ and $\lambda(p_{k+1}) < \lambda(p_k)$. All these operators satisfy $\|S_k - T\| < \varepsilon < \|T\|$, so they are nonzero and have norm less than L. Since there is no infinite strictly decreasing sequence of ordinals, after finitely many steps, say N, the peak is isolated or steep. Then Proposition 4.1, applied with $\eta = \varepsilon/2^{N+2}$, gives an operator in $\mathrm{NA}(X, Y)$ at distance less than $\varepsilon/2 + \sum_{k \ge 0} \varepsilon/2^{k+2} = \varepsilon$ from T. □

Remark 4.3. (1) The argument uses only the scalars ± 1; the complex case is not treated here. (2) The method needs isolated points at the bottom of the Cantor–Bendixson descent. It therefore says nothing about finite-dimensional spaces whose dual ball has a perfect set of extreme points.

5. Property scattered quasi-β

In this section Y is an arbitrary real Banach space, and we keep the notation $E = \operatorname{ext} B_{Y^*}$ and $K_Y = B_{Y^*}$. We recall property quasi-β [1], in the form stated in [6, Section 3.1]. A real Banach space Y has *property quasi-β* if there exist a subset $A \subseteq S_{Y^*}$, a mapping $\sigma\colon A \to S_Y$ and a function $\rho\colon A \to \mathbb{R}$ such that

(i) $f(\sigma(f)) = 1$ for every $f \in A$;
(ii) $|g(\sigma(f))| \le \rho(f) < 1$ whenever $f, g \in A$ and $f \neq g$;
(iii) for every $e \in E$ there exist a subset $A_e \subseteq A$ and $t \in \{1, -1\}$ such that $te \in \overline{A_e}^{w^*}$ and $\sup\{\rho(f) : f \in A_e\} < 1$.

If moreover $\rho(f) \le R$ for all $f \in A$ and some constant $R < 1$, then Y has Lindenstrauss' property β (in an equivalent formulation) [6].

5.1. **Definitions.**

Definition 5.1. Let Y be a real Banach space. An *admissible triple* for Y is a triple (A, σ, ρ) consisting of a subset $A \subseteq S_{Y^*}$, a mapping $\sigma\colon A \to S_Y$ and a function $\rho\colon A \to [0, 1)$ such that

(i) $f(\sigma(f)) = 1$ for every $f \in A$;
(ii) $|g(\sigma(f))| \le \rho(f)$ whenever $f, g \in A$ and $f \neq g$.

Definition 5.2. Let (A, σ, ρ) be an admissible triple for Y.

(a) An extreme point $e \in E$ is *good* if there exist a subset $A_e \subseteq A$ and $t \in \{1, -1\}$ such that $te \in \overline{A_e}^{w^*}$ and $\sup\{\rho(f) : f \in A_e\} < 1$.
(b) An extreme point $e \in E$ is *bad* if it is not good.

We write E_{good} and E_{bad} for the sets of good and of bad extreme points, so that $E = E_{\text{good}} \cup E_{\text{bad}}$ and $E_{\text{good}} \cap E_{\text{bad}} = \emptyset$. Both sets depend on the triple (A, σ, ρ).

Definition 5.3. A real Banach space Y has *property scattered quasi-β* if there exists an admissible triple (A, σ, ρ) for Y such that

(iii) for every $e \in E_{\text{bad}}$ there exists $t \in \{1, -1\}$ with $te \in \overline{A}^{w^*}$;
(iv) E_{bad} is scattered in the relative weak* topology, that is, every nonempty subset of E_{bad} contains a point which is isolated in that subset.

Remark 5.4. (1) Condition (i) says that f attains its norm at $\sigma(f)$; equivalently, by James' characterisation [5], $\sigma(f) \perp_B y$ for every $y \in \ker f$. Condition (ii) says that $\sigma(f)$ is a private norming point of f: every other member of A stays at height at most $\rho(f) < 1$ at $\sigma(f)$. In particular

$$\|f - g\| \ge f(\sigma(f)) - g(\sigma(f)) \ge 1 - \rho(f) \qquad (f, g \in A,\ f \neq g),$$

so $\rho(f)$ measures how closely other members of A may approach f.

(2) A good extreme point is automatically approached by A, since $\overline{A_e}^{w^*} \subseteq \overline{A}^{w^*}$; this is why condition (iii) is required only for bad points.

(3) A space has property quasi-β if and only if it has an admissible triple for which $E_{\text{bad}} = \emptyset$. Hence property quasi-$\beta$ implies property scattered quasi-β. Property β asks for ρ to be bounded away from 1 on all of A; property quasi-β asks this only near each extreme point; property scattered quasi-β asks it only near each extreme point outside a scattered set.

5.2. **Finite dimensions.**

Lemma 5.5. *Let Y be finite-dimensional, let (A, σ, ρ) be an admissible triple for Y satisfying condition* (iii) *of Definition 5.3, and let $e \in E_{\text{good}}$, with A_e and t as in Definition 5.2 and $\bar{\rho} = \sup\{\rho(f) : f \in A_e\} < 1$. Then $te \in A$ and $\rho(te) \le \bar{\rho}$. Moreover, no extreme point $e' \neq e$ satisfies $\|e' - e\| < 1 - \bar{\rho}$. In particular, every good extreme point is isolated in E.*

Proof. If $t = -1$, then $-e$ is an extreme point which is good with the same set A_e and the sign 1; and since $E = -E$ and $\|e' - e\| = \|(-e') - (-e)\|$, the conclusion for $-e$ implies the conclusion for e. So we may assume $t = 1$.

Since the weak* and norm topologies of Y^* coincide, there are $f_n \in A_e$ with $f_n \to e$ in norm. Suppose that infinitely many of the f_n are distinct; passing to a subsequence, they are pairwise distinct and $\sigma(f_n) \to y_1$ for some $y_1 \in B_Y$. By (ii), $|f_k(\sigma(f_n))| \le \rho(f_n) \le \bar{\rho}$ for $k \neq n$; letting $k \to \infty$ gives $|e(\sigma(f_n))| \le \bar{\rho}$, hence $|e(y_1)| \le \bar{\rho}$. On the other hand

$$|1 - e(\sigma(f_n))| = |(f_n - e)(\sigma(f_n))| \le \|f_n - e\| \to 0,$$

so $e(y_1) = 1$, a contradiction. Hence some f_n is repeated infinitely often, and it equals the limit e. So $e \in A$ and $\rho(e) \le \bar{\rho}$.

Put $y_e = \sigma(e)$. Let $e' \in E$ with $e' \neq e$ and $\|e' - e\| < 1 - \bar{\rho}$. Then $e'(y_e) \ge e(y_e) - \|e' - e\| > \bar{\rho}$. If e' is good, the first part gives $t'e' \in A$ for some $t' \in \{1, -1\}$, and $t'e' \neq e$ (if $t' = 1$ because $e' \neq e$; if $t' = -1$ because $\|e + e'\| \ge 2 - \|e' - e\| > 1$). If e' is bad, condition (iii) gives $t'e' \in \overline{A}$ for some $t' \in \{1, -1\}$; as $t'e' \neq e$ by the same reasoning, we may choose $f \in A$ so close to $t'e'$ that $f \neq e$ and $|f(y_e)| > \bar{\rho}$. In both cases we obtain $f \in A \setminus \{e\}$ with $|f(y_e)| > \bar{\rho} \ge \rho(e)$, which contradicts (ii). □

Proposition 5.6. *A finite-dimensional real Banach space Y has property scattered quasi-β if and only if* $\operatorname{ext} B_{Y^*}$ *is countable.*

Proof. Suppose that Y has property scattered quasi-β, witnessed by (A, σ, ρ). By Lemma 5.5, E_{good} consists of isolated points of the separable metric space E, so E_{good} is countable. We show that the scattered set E_{bad} is countable. If not, let C be the set of points $e \in E_{\text{bad}}$ every neighbourhood of which contains uncountably many points of E_{bad}. Since E is second countable, $E_{\text{bad}} \setminus C$ is countable, so C is uncountable. If $e \in C$ and U is a neighbourhood of e, then $U \cap E_{\text{bad}}$ is uncountable while $E_{\text{bad}} \setminus C$ is countable, so U contains a point of C other than e. Thus C is a nonempty subset of E_{bad} without isolated points, contradicting (iv). Hence $E = E_{\text{good}} \cup E_{\text{bad}}$ is countable.

Conversely, suppose that E is countable. By Lemma 2.7, E is scattered, so its isolated points are dense in E. Every isolated point of E is exposed, because exposed points are dense in E by Straszewicz's theorem [14]. Let A contain exactly one point of each pair $\pm e$ of isolated points of E. For $f \in A$, let $\sigma(f) \in Y$ expose f, normalised so that $f(\sigma(f)) = 1$; then $h(\sigma(f)) < 1$ for every $h \in K_Y \setminus \{f\}$, and $\|\sigma(f)\| = \max_{h \in K_Y} h(\sigma(f)) = 1$. Since f and $-f$ are isolated in E and $-f \notin A$, the closure of $A \setminus \{f\}$ is a compact subset of $K_Y \setminus \{f, -f\}$. For h in this compact set, $h(\sigma(f)) < 1$, and $-h(\sigma(f)) < 1$ because $-h \in K_Y \setminus \{f\}$. Hence

$$\rho(f) = \sup\{|g(\sigma(f))| : g \in A \setminus \{f\}\} < 1$$

(put $\rho(f) = 0$ if $A = \{f\}$), and (A, σ, ρ) is an admissible triple. Every isolated point e of E is good, with $A_e = \{e, -e\} \cap A$. So E_{bad} is contained in the set of non-isolated points of E, and is scattered as a subset of the scattered set E; this is (iv). For (iii), let $e \in E_{\text{bad}}$ and choose isolated points $e_n \to e$. For each n, $e_n \in A$ or $-e_n \in A$; one of the two alternatives occurs for infinitely many n, which gives $e \in \overline{A}$ or $-e \in \overline{A}$. □

Corollary 5.7. *A finite-dimensional real Banach space Y has property quasi-β if and only if* $\operatorname{ext} B_{Y^*}$ *is discrete.*

Proof. If Y has property quasi-β, take a triple with $E_{\mathrm{bad}} = \emptyset$ (Remark 5.4(3)); condition (iii) of Definition 5.3 holds vacuously, so every extreme point is isolated by Lemma 5.5. Conversely, if E is discrete, then it is countable, and the triple constructed in the proof of Proposition 5.6 makes every point of E good, so $E_{\mathrm{bad}} = \emptyset$. □

This recovers the observation of [1, p. 414] recalled in [6, Remark 3.19].

Corollary 5.8. *Every finite-dimensional real Banach space with property scattered quasi-β has property B.*

Proof. Combine Proposition 5.6 and Theorem 1.1. □

Remark 5.9. Let Y be finite-dimensional and let (A, σ, ρ) satisfy (iii). By Lemma 5.5, an extreme point e is good if and only if $e \in A$ or $-e \in A$ (for the converse take $A_e = \{te\}$). Hence $E_{\mathrm{bad}} = E \setminus (A \cup -A)$: in finite dimensions the bad points are exactly the extreme points without a private norming point of their own.

5.3. **Examples.**

Example 5.10. Let $n \geq 2$, and let $\langle \cdot, \cdot \rangle$ and $\mathbf{e}_1, \dots, \mathbf{e}_n$ denote the Euclidean inner product and the standard unit vectors of $\mathbb{R}^n$. For $k \in \mathbb{N}$ put $\zeta_k = \cos(1/k)\,\mathbf{e}_1 + \sin(1/k)\,\mathbf{e}_2$, and let

$$\Gamma = \{\pm \mathbf{e}_j : 1 \leq j \leq n\} \cup \{\pm \zeta_k : k \in \mathbb{N}\}.$$

Let Y_n be $\mathbb{R}^n$ with the norm $\|x\| = \max\{|\langle \zeta, x \rangle| : \zeta \in \Gamma\}$. Then Y_n has property scattered quasi-β but fails property quasi-β.

Proof. The set Γ is compact (its only accumulation points are $\pm \mathbf{e}_1 \in \Gamma$), symmetric, spans $\mathbb{R}^n$, and lies on the Euclidean unit sphere. Hence the norm is well defined, and, identifying Y_n^* with $\mathbb{R}^n$ through $\langle \cdot, \cdot \rangle$, $B_{Y_n^*} = \operatorname{conv} \Gamma$ by the bipolar theorem. Every point of Γ lies on the Euclidean unit sphere, so it is an extreme point of the Euclidean unit ball, and hence of $\operatorname{conv} \Gamma$. Conversely, $\operatorname{ext} \operatorname{conv} \Gamma \subseteq \Gamma$ by Milman's theorem [11]. Thus $\operatorname{ext} B_{Y_n^*} = \Gamma$, which is countable, so Y_n has property scattered quasi-β by Proposition 5.6. Since $\zeta_k \to \mathbf{e}_1$ and $\zeta_k \neq \mathbf{e}_1$, the point $\mathbf{e}_1$ is not isolated in $\operatorname{ext} B_{Y_n^*}$, so Y_n fails property quasi-β by Corollary 5.7. □

Remark 5.11. The bad points may have any countable Cantor–Bendixson level. Every countable compact metric space embeds in $[0, 1]$ [8]. Given a countable ordinal α, choose a compact set Γ_0 in the arc $\{\cos\vartheta\,\mathbf{e}_1 + \sin\vartheta\,\mathbf{e}_2 : \pi/8 \leq \vartheta \leq 3\pi/8\}$ which is homeomorphic to the ordinal $\omega^\alpha + 1$, and replace $\{\zeta_k\}$ by Γ_0 in Example 5.10. The same proof shows that $\operatorname{ext} B_{Y^*} = \Gamma_0 \cup -\Gamma_0 \cup \{\pm \mathbf{e}_j\}$. The set Γ_0 is relatively open and closed in this set, so it contains a point of level α. Thus property quasi-β corresponds to level 0 only, while property scattered quasi-β allows every countable level; in particular the levels met in the descent in the proof of Theorem 1.1 may be infinite ordinals, although the descent itself always has finitely many steps.

Remark 5.12. In dimension two, the set of extreme points of a compact convex set is closed, so properties β and quasi-β both coincide with polyhedrality (by Corollary 5.7, and because a polyhedral space has property β [9]). In particular Example 5.10 with $n = 2$ is not polyhedral and has property B by Theorem 1.1. From dimension three on, property quasi-β is strictly weaker than polyhedrality. For instance, let $Q \subseteq \mathbb{R}^3$ be the closed convex hull of $\pm(1, 0, 1)$, $\pm(1, 0, -1)$ and $\pm(\cos(1/k), \sin(1/k), 0)$, $k \in \mathbb{N}$. Projecting onto the first two coordinates, all these points lie over the unit circle, and each point $\pm(\cos(1/k), \sin(1/k), 0)$ is the only point

of Q over its projection, while the points of Q over $\pm(1,0)$ form the segments with endpoints $\pm(1,0,\pm1)$. Hence all the listed points are extreme points of Q. By Milman's theorem the only other candidates are $\pm(1,0,0)$, the midpoints of these segments, which are not extreme. So $\operatorname{ext} Q$ is infinite and discrete, and the space whose dual ball is Q has property quasi-β by Corollary 5.7 without being polyhedral.

Question 5.13. Does every real Banach space with property scattered quasi-β have property B?

In infinite dimensions, countability and scatteredness of E_{bad} are no longer equivalent conditions, and the proof of Theorem 1.1 does not apply as it stands: the landing set lies in Y^{**}, approach directions are only weak* limits, and the uniform gap of Definition 2.2 requires Y^* to have the Radon–Nikodým property.

6. Contributions of the witness and steerer

This section records, as accurately as the written record allows, the contributions of Debmalya Sain to this manuscript. It is based on the transcripts of the conversations between D. Sain and the author between 24 and 28 October 2026, and on the running research record kept during that work. We distinguish between ideas and questions that originated with D. Sain, decisions he made about the direction of the work, and his interventions that changed the mathematics. We have tried neither to understate nor to overstate his role.

Ideas and questions that originated with D. Sain.

(1) *The project and the first strategy.* The investigation was proposed by D. Sain as "A quest towards Property B", an exploration of which finite-dimensional spaces have Lindenstrauss property B. His original expectation was that a finite-dimensional space whose dual ball has only isolated extreme points has property B, with the following strategy: first make the adjoint absolutely strongly exposing, then change it only at its peak, using the Bishop–Phelps theorem. That statement turned out to be covered by the theorem of Acosta, Aguirre and Payá, but the strategy survived: it is the unique-peak step (Corollary 2.3) and the finishing step at isolated peaks (Case 1 of Proposition 4.1) of the proof of Theorem 1.1.

(2) *Preservation of orthogonality as a tool.* D. Sain maintained, at a time when the author had underrated it, that his theorem on the preservation of Birkhoff–James orthogonality at points of norm attainment [12, Theorem 2.3] would be a key tool. It became the conceptual source of the floor (Lemma 2.4) and of Lemma 2.5, on which the flat case of the proof rests; see Remark 2.6 for the precise logical relation.

(3) *The estimate at non-smooth peaks.* D. Sain proposed an estimate for the adjoint at a non-smooth point f of the dual ball, in terms of the diameter of the face M_f. This estimate led the author to the per-operator landing points, which became the landing set W_S and the flat-or-steep dichotomy (Definition 3.2).

(4) *Property scattered quasi-β.* D. Sain proposed the new property as a minimal change of property quasi-β: the conditions of Acosta, Aguirre and Payá are kept verbatim, and only the uniformity near extreme points is allowed to fail, on a scattered set of bad extreme points. He also proposed framing Theorem 1.1 as a generalisation of properties β and quasi-β (Section 5).

(5) *Examples.* D. Sain asked for an n-dimensional space with property scattered quasi-β but without property quasi-β, to give the new definition more legitimacy; this led to Example 5.10 and Remark 5.11.

(6) *Connection with known questions.* D. Sain insisted that the work be related to the published question of Jung, Martín and Rueda Zoca [6, Remark 3.24], which Corollary 1.2 answers.

Decisions about the direction of the work.

(7) D. Sain fixed the working method: a first phase of exploration, pushed as far as possible, with honest and critical evaluation at every step, followed by a second phase in which he checks and understands everything. He asked the author to be more ambitious in the exploration.
(8) D. Sain decided to concentrate on finite-dimensional spaces, which is the setting of Theorem 1.1.

Interventions that changed the mathematics.

(9) D. Sain pointed out that an earlier version of Definition 5.2 defined only bad points, and not good points. This led to the present logical order: admissible triple, then good and bad points, then the property.
(10) D. Sain required complete definitions written in his style and the rule "one object, one symbol" throughout the manuscript, which led to a complete revision of the notation.
(11) D. Sain required that every result used be acknowledged, and asked for the definitions of the literature to be checked against the original sources.

What the witness and steerer did not do. D. Sain did not write any of the proofs in this manuscript, and at the time of writing he has not completed a verification of them. The improvements made for this version (Definition 2.2, Lemma 2.5, the extension of Corollary 1.2 to $C(K)$ for countable K, and Corollary 5.7) were made by the author and have not been seen by him in this form before this manuscript. Responsibility for every proof, and for any error, rests with the author.

7. How this manuscript was produced

The results, proofs, examples and text of this manuscript were produced by the author, CLAUDE, in conversation with D. Sain, as described in Section 6. The proof of Theorem 1.1 was checked by the author and, independently, by a separate instance of the same AI model acting as a referee (25 October 2026); for this version every proof was checked again, line by line, by the author (28 October 2026). No human has verified the proofs.

The definitions of properties quasi-β and β, Proposition 1.1(a), Example 3.18, Remark 3.19 and Remark 3.24 of [6] were checked against the arXiv version of that paper. The literature was searched for earlier proofs of Theorem 1.1 or of Corollary 1.2, and none was found; the search consisted of web searches only and was not exhaustive. In particular, the authors of [6] and of the related literature on ACK structures should be consulted before Theorem 1.1 is regarded as new.

The following references have not been verified against the original sources by the author, and should be checked: the bibliographic details of [1], [7], [10] and [14], and the volume and article number of [6].

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Author: CLAUDE (Claude Opus 5.5), an AI model developed by Anthropic.

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